

How would your code scale up with this larger dataset? Would you need any additional computing resources? Be as specific as possible.

My code should scale well with the larger dataset. The time it takes to run each line/dplyr function will increase. If necessary, the code can be formatted using a *Data Table* which was created to work with large datasets.

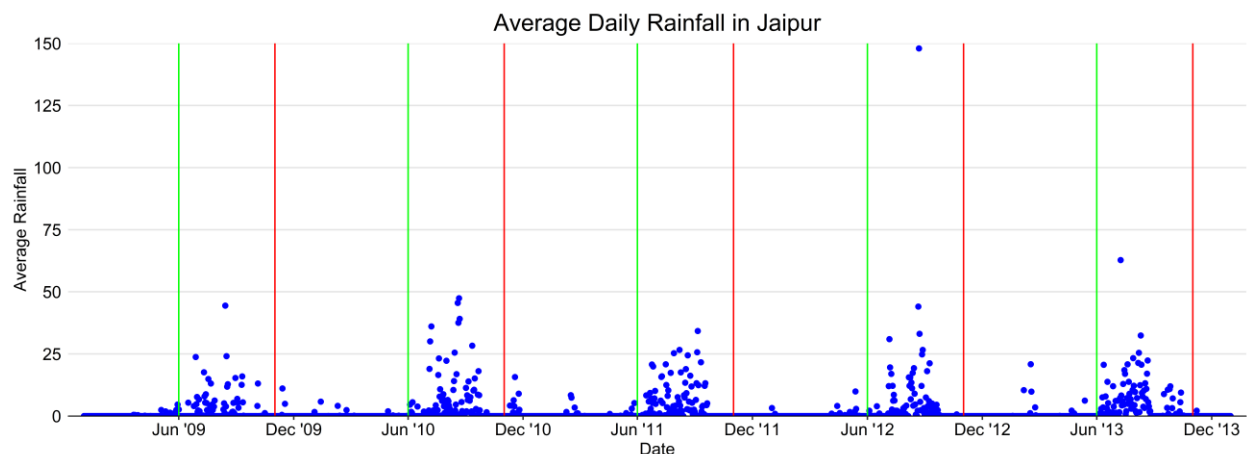
Ideally, for larger data sets I would have access to a computing cluster via school, work, or an Amazon Web Services (AWS) subscription. My laptop has limited memory and disk space, thus the code takes a while to run and easily overloads my computer causing the R to crash. I use the Linux cluster at the Francisco Federal Reserve Bank of San Francisco to run computationally heavy tasks.

1. For the Jaipur district in Rajasthan, create a time series dataset of daily rainfall, averaged over the five years.

Saved as .csv file *jaipur.csv*.

2. Using this time series, create a scatterplot of rainfall by day. On what day does the monsoon season start in Jaipur? When does it end? Indicate your answer on the graph.

Monsoon season starts in early June (green) and ends in late September (red).



3. Format your graphs so that they are publication-quality and save them as .pdf files.

Save as *"3_Jaipur_Monsoon.pdf"*

4. Create a publication-quality table of annual average temperature by state and year.

**Table 1. Annual Average Temperature
by State & Year**

State	Year	Average Temperature (°C)
Gujarat	2009	27.54
Gujarat	2010	27.49
Gujarat	2011	26.94
Gujarat	2012	26.62
Gujarat	2013	26.59
Kerala	2009	25.76
Kerala	2010	25.91
Kerala	2011	25.53
Kerala	2012	25.83
Kerala	2013	25.79
Pondicherry	2009	27.87
Pondicherry	2010	27.80
Pondicherry	2011	27.49
Pondicherry	2012	27.87
Pondicherry	2013	27.75
Punjab	2009	23.49
Punjab	2010	23.59
Punjab	2011	22.88
Punjab	2012	22.88
Punjab	2013	22.99
Rajasthan	2009	26.75
Rajasthan	2010	26.68
Rajasthan	2011	25.86
Rajasthan	2012	25.60
Rajasthan	2013	25.62

We are interested in using this dataset to estimate the impact of temperature on yearly district-level mortality in India.

1. Write out a simple econometric model regressing mortality on temperature.

$$Mortality_{i,t} = \alpha + \phi_1 Temperature_{i,t} + \varepsilon_{i,t}$$

2. Suppose you estimate the model using OLS and obtain a coefficient of 0.05 with a standard error of 0.02. Interpret this result.

The positive coefficient suggests that there is a positive relationship between higher temperatures and mortality rates. As temperature rises, so will mortality. All else equal, a one unit Celsius increase is correlated with an increase in mortality rates by 0.05. Dividing the coefficient by the standard error we estimate a t-value of 2.5

3. The effect of temperature on mortality is very likely nonlinear. How would you modify your model above to estimate any potential nonlinearities?

I would modify the model logging the left-hand side of the regression to make a linear-log regression function. This means that a one unit change in Celsius can be interpreted as a one unit change in Celsius causing an X% change in mortality. This method can capture some potential nonlinearities.

4. Is your parameter of interest identified? How could you use a) fixed effects and b) other control variables to remove potentially confounding factors?

My parameter of interest is mortality rates.

Year fixed effects should be used to account for variation across years in the dataset. State fixed effects help control for difference across states. Both account for changes across years or states in order to account for changes that occur across years or states, but are the same within. A control for the population's age is warranted as mortality rates are highly correlated with age. Older districts may have higher rates due to their aging population and not temperature effects. Income is another control that can attempt to control for factors, such as better access to healthcare, insurance, and nutrients/diets for example. Education level is correlated with income and mortality, thus it is possible that it be used with an interaction term. Interactions are necessary when the variable affects both dependent variable and an independent variable in the regressions.

5. Write out a full econometric model addressing the concerns raised in 3 and 4.

$$Mortality_{i,t} = \alpha + \varphi_1 Temperature_{i,t} + Age_{it} + Education_{it} + Income_{i,t} + \varepsilon_{i,t}$$

The standard error can be used by estimating the adjusted-R² value. This explains the percentage of the variable of interest that is explained in the model. A variance decomposition can be used to isolate each explanatory's average effect at the state level.